

Francesco Dettori — PhD Student

 Francesco Dettori |  francesco.dettori@inria.fr |  +33 07 49 11 39 10

SUMMARY

Biomedical engineer and PhD student at Inria Strasbourg, working on patient-specific simulation and digital twin technologies for liver tumour ablation. My research combines medical imaging, numerical modelling and clinical validation, with a strong interest in translating engineering solutions into practical healthcare innovations.

WORK EXPERIENCE

PhD Student - INRIA, team Mimesis, Strsbourg

May 2025 - Present

Developing patient-specific simulation tools for liver tumor thermal ablation within the [MediTwin project](#), combining medical imaging, computational modeling and experimental validation. Collaborating with clinicians and engineers to personalize model parameters, evaluate simulations using preclinical data and explore their integration into treatment planning.

Research Intern - INRIA, team Mimesis, Strsbourg

Feb 2024 - Aug 2024

Developed deep-learning methods for real-time 3D tracking of endovascular devices using Fiber Bragg Grating sensors, contributing to research on autonomous robotic catheter navigation.

PUBLICATIONS

- 2026 **F. Dettori**, I. Nahmed, M. Duprez, J. Verde, P. Alvarez, S. Cotin. *Personalization of Liver Microwave Ablation Simulation*. International Journal of Computer Assisted Radiology and Surgery. Presented at IPCAI 2026 – Best Presentation Audience Choice Award.
- 2026 I. Nahmed, **F. Dettori**, J. Verde, M. Duprez, P. Alvarez, S. Cotin. *A Digital Twin for Microwave Liver Treatment Re-planning*. International Journal of Computer Assisted Radiology and Surgery.
- 2026 **F. Dettori**, I. Nahmed, M. Duprez, P. Alvarez, S. Cotin. *Real-Time Interactive MWA Planning via Probabilistic Inverse Inference*. Accepted at MICCAI 2026, to appear in Springer LNCS.
- 2026 I. Nahmed, **F. Dettori**, M. Duprez, P. Alvarez, S. Cotin. *A Deep Learning Surrogate Model for Microwave Thermal Ablation: From Preoperative Planning to Intraoperative Re-planning*. Accepted at MICCAI 2026, to appear in Springer LNCS.

EDUCATION

- 2022 - 2025 MSc, Biomedical Engineering at **Politecnico di Milano** (Grade 110/110 cum laude)
- 2023 - 2024 HealthTech Master Program at **University of Strasbourg** (Mention: Très Bien)
- 2019 - 2022 BSc, Ingegneria Biomedica at **Politecnico di Milano** (Grade: 110/110)

OTHER PROJECTS

AI in Radiological Images for PCNSL

[Report](#)

Used Deep Learning on MRI data to predict life expectancy for PCNSL patients, aiding in tailored treatments. (*Bachelor thesis project*)

Business Plan for "Budd-e" Robot

[Report](#)

Created a business plan for a self-driving robot assisting visually impaired individuals, including strategic and financial analyses. (*Management of organizations and projects course*)

Opacimeter Development

[Report](#)

Developed an Arduino-based device to measure material opacity, applicable in various industries. (*Technologies for Sensors* course)

Brembo S.p.A. Analysis

[Report \(ITA\)](#)

Analyzed Brembo's strategic direction and financial performance for 2020/21. (*Economia e Organizzazione Aziendale* course)

Mood Disorder Monitoring System

[Report](#)

Developed a system using Microsoft Access for continuous patient monitoring and therapy management. (*Medical Informatics* course)

Femoral Stem CAD Model

[Demo](#)

Replicated a femoral stem used for hip replacements using SolidWorks. (*Biomechanics* course)

Wearable Distance Sensor for Blind People

Developed an Arduino-based wearable sensor during a hackathon to aid visually impaired navigation with customizable feedback.

SKILLS

Programming	Python, MATLAB, C, SQL, R
Tools & Frameworks	PyTorch, SimpleITK, 3D Slicer, Git
Research Methods	Medical image analysis, deep learning, numerical simulation, statistical analysis
Languages	Italian (Native), English (Professional), French (Base)